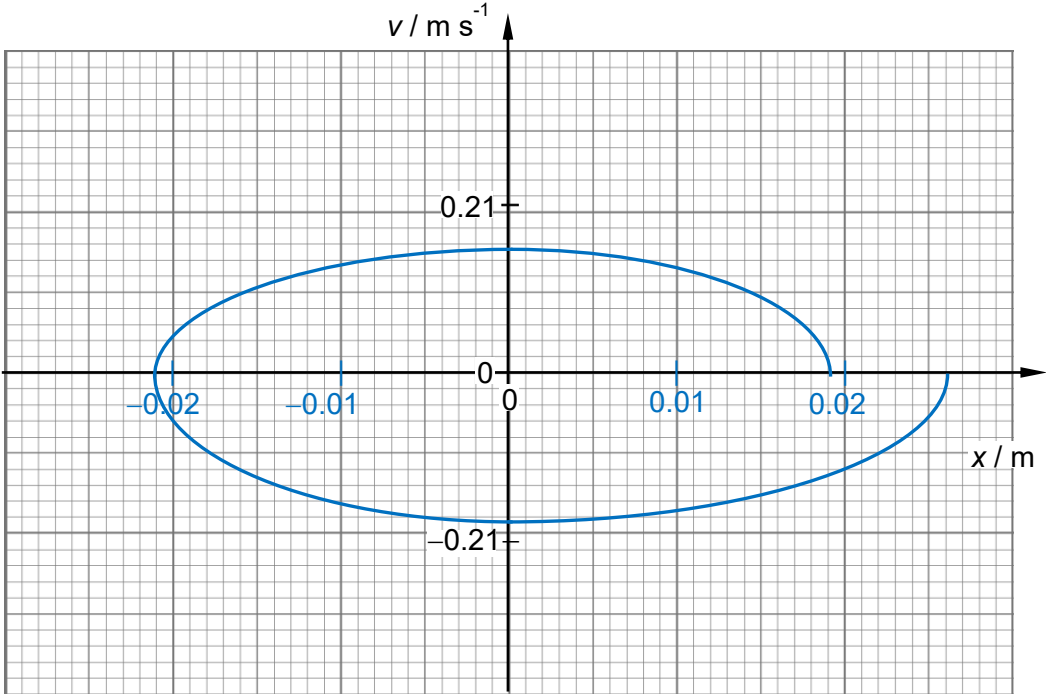
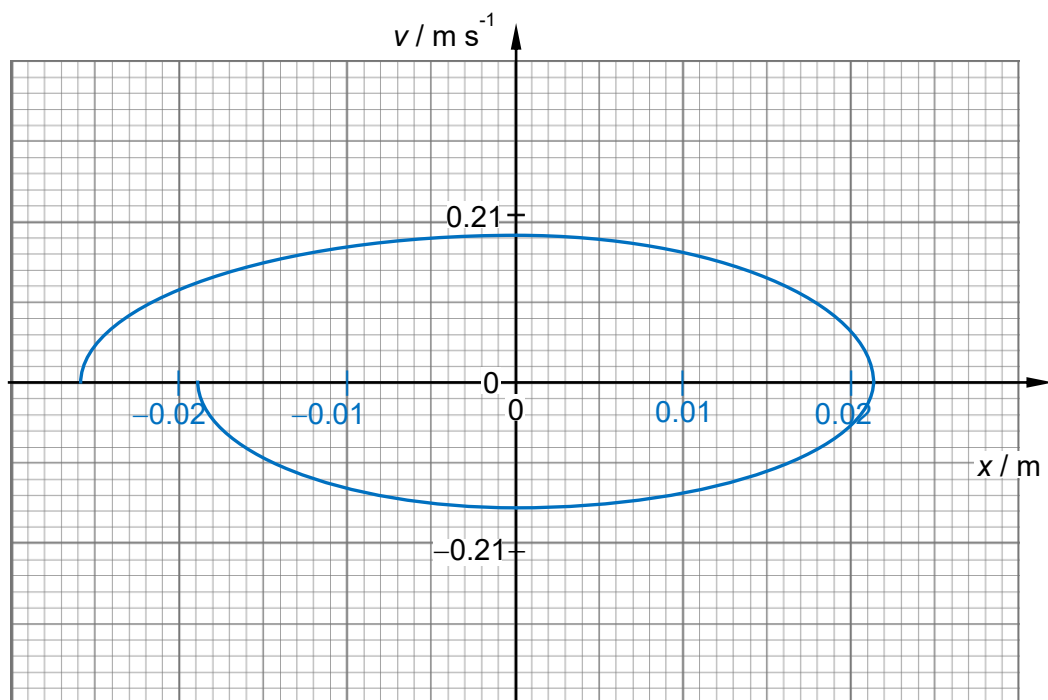


Q8		
(a)	The net external force acting on a body is zero and the net moment on the body about any point is zero.	
(b)	(i) $T \sin \theta = kx$ $k = \frac{T \sin \theta}{x} = \frac{2.50 \sin 36^\circ}{0.050}$ $= 29.389$ $= 29.4 \text{ N m}^{-1} \text{ or } 29 \text{ N m}^{-1} \text{ (to 2 s.f.)}$	
	(ii) $T \cos \theta = mg$ $m = \frac{T \cos \theta}{g} = \frac{2.50 \cos 36^\circ}{9.81}$ $= 0.206 \text{ kg or } 0.21 \text{ kg (to 2 s.f.)}$	
	(iii) $\text{EPE} = \frac{1}{2} kx^2 = \frac{1}{2} (29.4)(0.050)^2$ $= 0.0368$ $= 0.037 \text{ J}$	
(c)	(i) Simple harmonic motion is a periodic motion in which the acceleration of the body is directly proportional to the displacement from its equilibrium point, and is always in the opposite direction to the displacement (or always directed to equilibrium position.)	
	(ii)1. By conservation of energy, Loss in GPE = Gain in KE $mgh = \frac{1}{2} mv_{\max}^2$ $v_{\max} = \sqrt{2gh} = \sqrt{2g(L - L \cos \theta)}$ $= \sqrt{2(9.81)(0.150 - 0.150 \cos 10^\circ)}$ $= 0.21145$ $= 0.21 \text{ ms}^{-1}$	
	(ii)2. T will be greater than W because, in addition to <u>supporting the bob's weight</u> , the string must <u>provide the necessary centripetal force to keep the bob moving in its circular path.</u> OR <u>The vector sum of the weight and tension must provide the centripetal force for circular motion pointing upwards at the bottom.</u>	

	(ii)3. T – W provides the centripetal force for the bob to move in its circular arc $T - W = \frac{mv^2}{r}$ $T = \frac{mv^2}{r} + mg = \frac{0.206(0.211^2)}{0.15} + 0.206(9.81)$ $= 2.082 \text{ N}$ $= 2.1 \text{ N}$	
(d)	(i) Correct shape starting at 0.026 m (reasonable smooth spiral) Correct labelling of $v_{\max}$ and x_0 $x_0 = L \sin \theta = 0.15 \sin 10^\circ = 0.02604 = 0.026 \text{ m}$  OR	

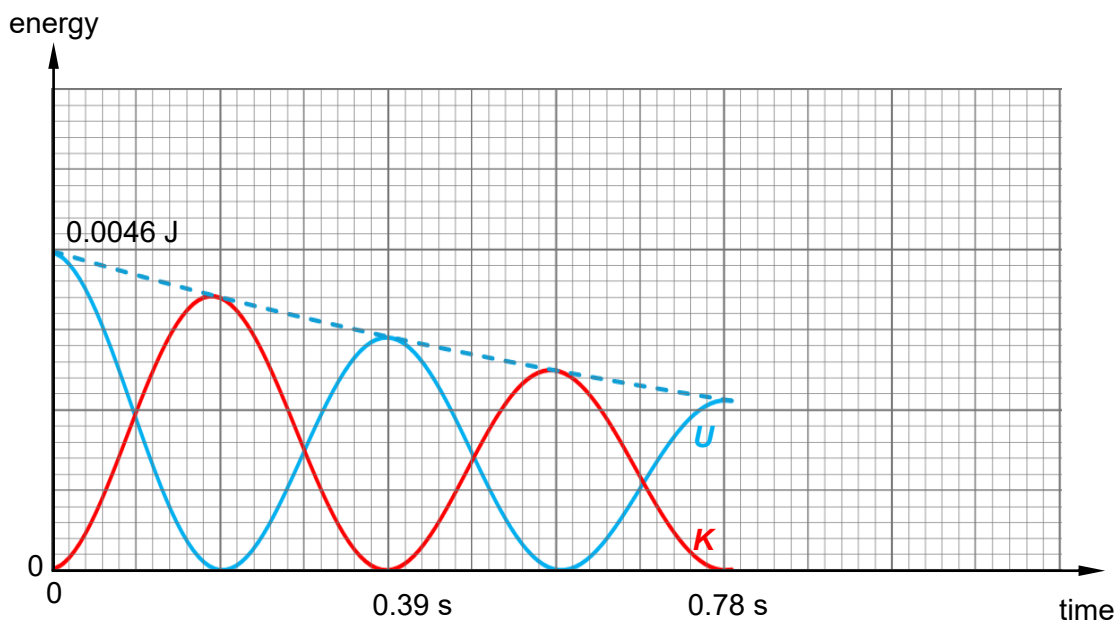


(ii)

$$v = \omega x_0 = \left(\frac{2\pi}{T} \right) x_0$$

$$T = \frac{2\pi x_0}{v} = \frac{2\pi(0.026)}{0.21} = 0.78 \text{ s}$$

$$\text{Total initial energy} = \frac{1}{2}mv^2 = \frac{1}{2}(0.21)(0.21)^2 = 0.0046 \text{ J}$$



Correct period labelled (2 cycles)

Correct shape of U Correct shape of K

